Network Port Scanner

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PROBLEM STATEMENT

- **Problem:**
- Network administrators and security professionals often need to discover which ports are open on a target system. Manual port checking is time-consuming, error-prone, and lacks a user-friendly interface.
- **Need:**
- A fast, reliable, and visually intuitive GUI-based tool that can scan a range of TCP ports on any IP address or hostname, display open ports with service names, and show real-time scan progress.

Project Description

The Network Port Scanner is a Python-based desktop application with a minimal GUI built using Tkinter. It allows users to scan any IP address or hostname for open TCP ports within a specified range.

The tool uses multi-threading for fast parallel scanning and displays results in real time. It identifies open ports along with their common service names (e.g., Port 53 - DNS, Port 443 - HTTPS, Port 853 - DoT).

A progress bar shows scan completion status with elapsed time. The application demonstrated scanning 8.8.8.8 (Google DNS), successfully identifying 3 open ports in the range 1-1024.

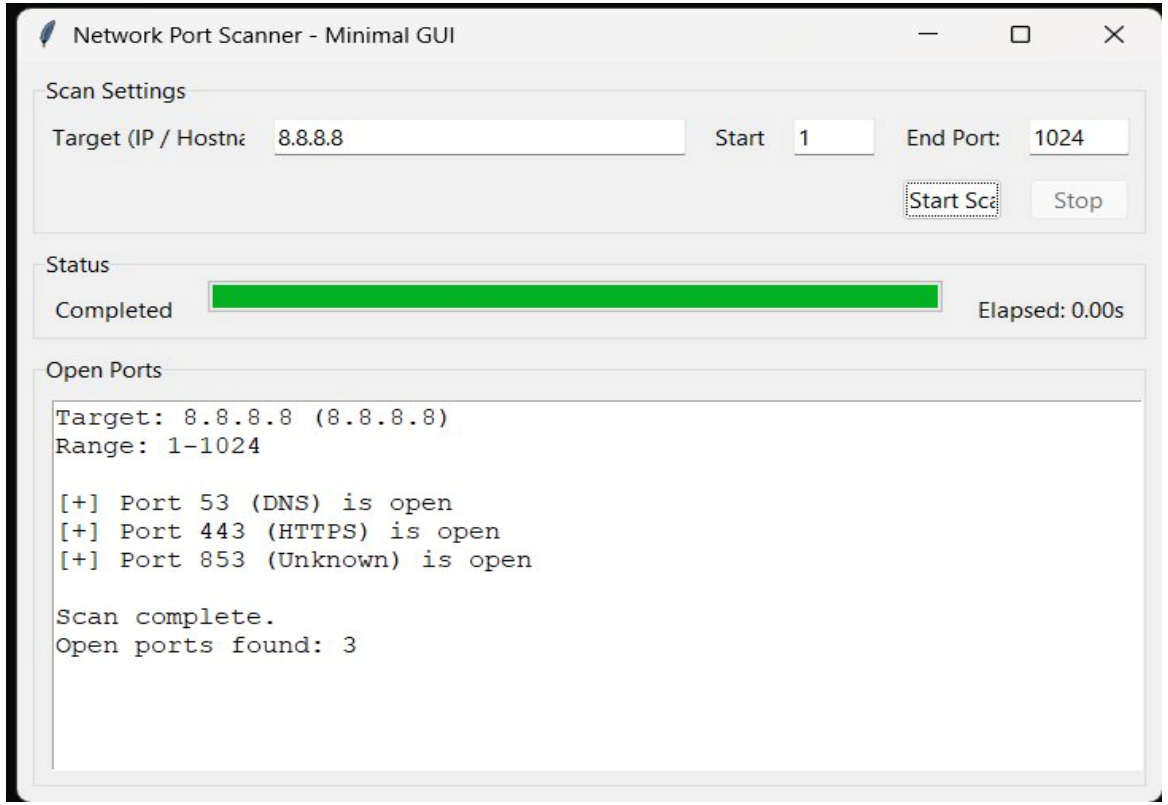
WHO ARE THE END USERS?

- **Network Security Professionals**
- Pen testers and ethical hackers who need to identify exposed services on target systems.
- **System & Network Administrators**
- IT staff who audit their own infrastructure to detect unauthorized open ports and reduce attack surface.
- **Students & Researchers**
- Cybersecurity learners who want to understand TCP/IP, port scanning concepts, and network reconnaissance in a hands-on way.

Technology Used

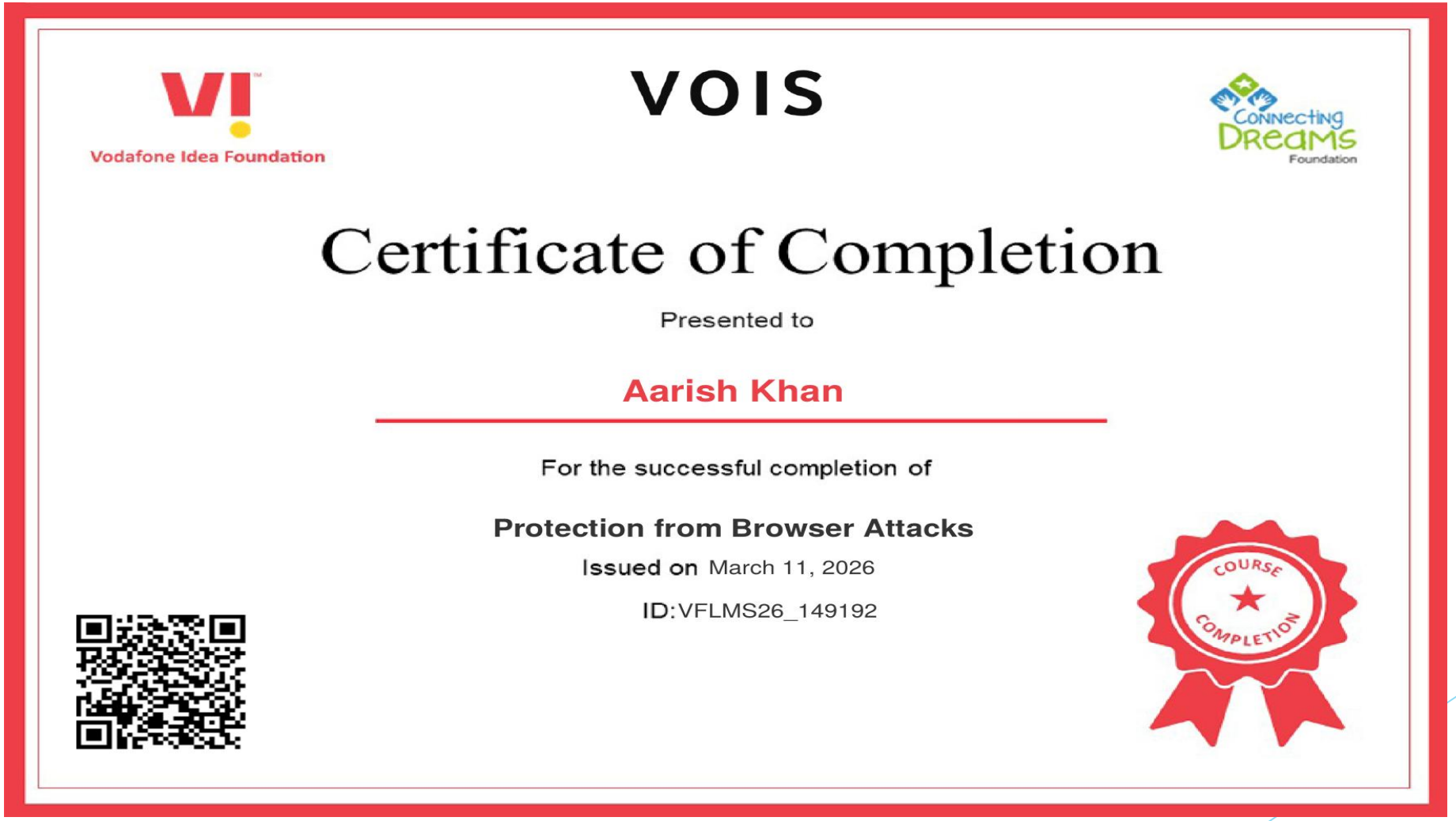
- **Python 3** - Core programming language for the application logic
- **Tkinter** - GUI framework for the desktop interface (buttons, progress bar, text area)
- **Socket Library** - Python built-in for TCP connection attempts and hostname resolution
- **Threading Library** - Multi-threaded scanning for faster port discovery
- **OS: Windows 10/11** - Development and testing environment

RESULTS



Demo Link

Certificate - Protection from browser attacks



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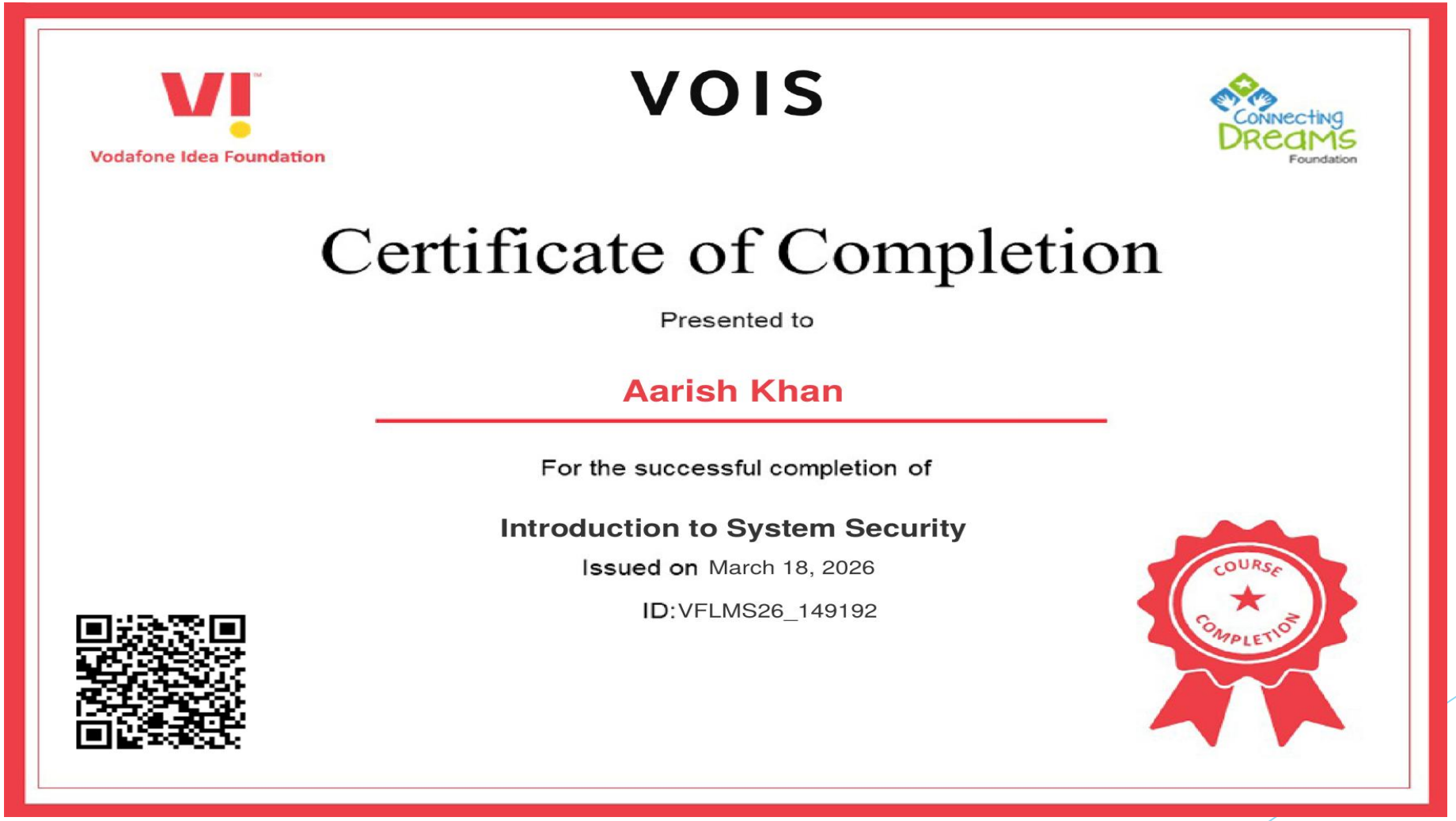

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GitHub link: https://github.com/aarishkhan/network-port-scanner

Thank you